

Math 2551 Worksheet: Optimization I

1. Find all the local maxima, local minima, and saddle points of $f(x, y) = e^y(x^2 - y^2)$.
2. Find and classify all critical points for the function $x^3 + 3xy + y^3$.
3. Can you conclude anything about $f(a, b)$, if f and its first and second partial derivatives are continuous around the critical point (a, b) and $f_{xx}(a, b)$ and $f_{yy}(a, b)$ differ in sign? Justify your answer.
4. In each case, the origin is a critical point of f and $f_{xx}f_{yy} - (f_{xy})^2 = 0$ at the origin, so the Second Derivative Test fails at the origin. Use some other method to determine whether the function f has a maximum, a minimum, or neither at the origin.
 - (a) $f(x, y) = x^2y^2$
 - (b) $f(x, y) = 1 - x^2y^2$
 - (c) $f(x, y) = xy^2$
 - (d) $f(x, y) = x^3y^2$
 - (e) $f(x, y) = x^3y^3$
 - (f) $f(x, y) = x^4y^4$